

# RayCastED: A RayCast End-to-end Detection Model for Lightweight and Boundary-Aware Nuclei Detection

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# Clinical Background: TILs Assessment

- **Tumor-infiltrating lymphocytes (TILs)** are established prognostic biomarkers in multiple cancers
- International TILs Working Group guidelines recommend **area-based scoring**: percentage of stromal area occupied by lymphocytes
- **Critical requirement**: accurate boundary delineation, not just centroid detection

## The Challenge

Manual assessment is:

- Time-consuming
- Subject to inter-observer variability
- Impractical for whole-slide images

$$\text{sTILs (\%)} = \frac{\text{Area of mononuclear inflammatory cells in stroma}}{\text{Total intratumoral stromal area}} \times 100$$

# Clinical Background: Immune Cell Density Scoring

- **Immune Cell Density** quantifies the absolute number of immune cells within a defined tissue region.
- Represents a shift from traditional visual area estimation to precise **single-cell enumeration**.
- **Critical requirement:** robust **instance segmentation** to accurately detect, separate, and classify individual nuclei.

## The Challenge

Manual single-cell counting is:

- Practically impossible at scale
- Highly prone to human error
- Requires digital AI pipelines for whole-slide analysis

$$\text{Immune Cell Density} = \frac{\text{Total count of detected immune cells}}{\text{Analyzed tissue area (mm}^2\text{)}}$$

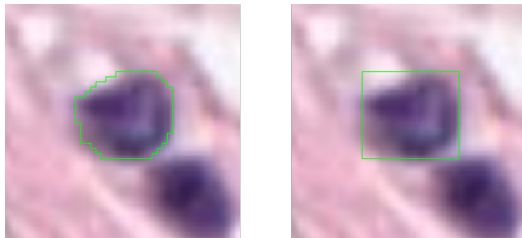
# The Clumped Cell Problem

- Densely clustered nuclei merge in segmentation masks
- **Result:** Cell undercounting
- Existing solutions require costly post-processing
- Non-Max Suppression (NMS) suppresses overlapping predictions (risky in clumped regions)

## Key Issues

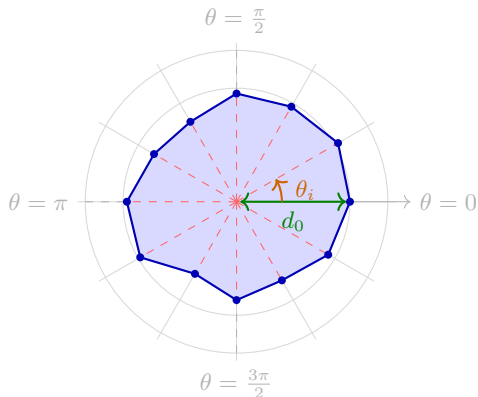
- 1 Instance separation overhead
- 2 NMS-based suppression errors
- 3 Deployment inaccessibility (transformers)

# The Accuracy-Deployment Trade-off



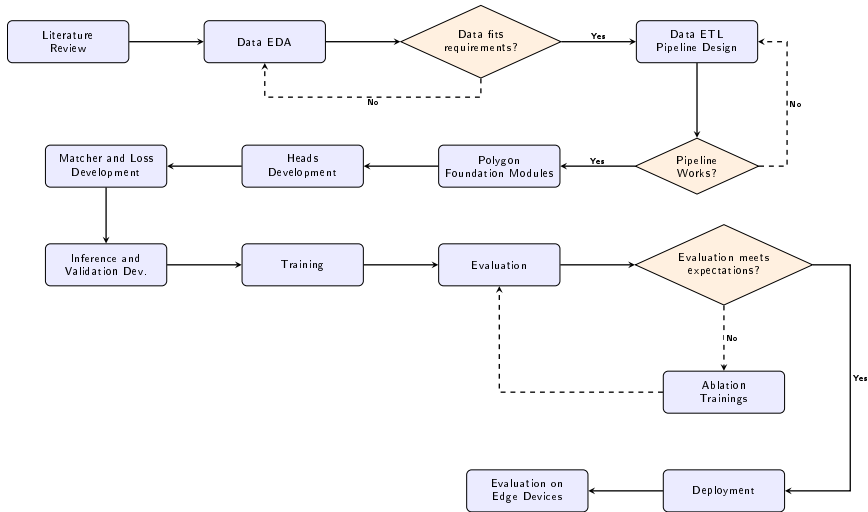
| Approach                                     | Pros                                  | Cons                                                   |
|----------------------------------------------|---------------------------------------|--------------------------------------------------------|
| Instance Segmentation<br>(U-Net, Mask R-CNN) | Accurate boundaries                   | Computationally expensive<br>(unsuitable for edge)     |
| Transformer-based<br>(CellViT, DINO)         | Strong performance<br>Good separation | Quadratic complexity<br>(memory/compute intensive)     |
| Object Detection<br>(YOLO, Faster R-CNN)     | Fast, lightweight<br>Edge-deployable  | <b>No boundary geometry</b><br>(insufficient for TILs) |

# Star-Convex Polygons for Cell Representation



- **Left:** Polar coordinate representation with centroid  $\mathbf{c}$  and radial distances  $d_i$
- **Right:** Applied to real cell with 12 rays at fixed angles  $\theta_i$
- Preserves boundary geometry while maintaining per-cell separation

# Research Workflow



# Datasets

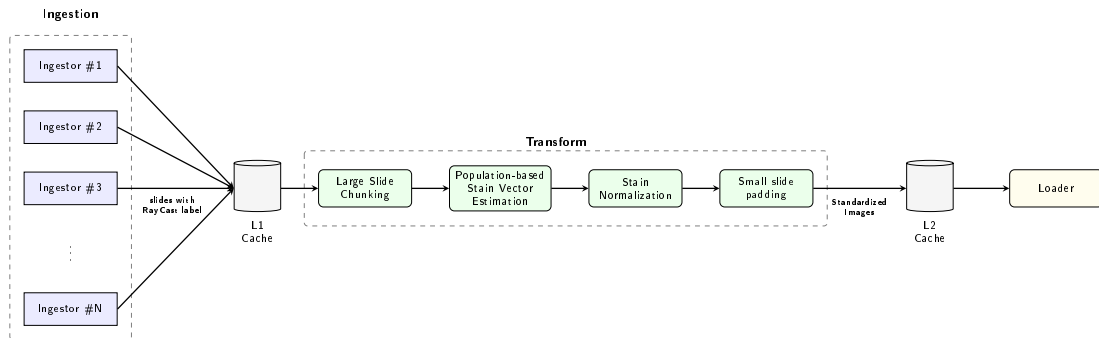
| Dataset   | Purpose              | Tissue      | Format        |
|-----------|----------------------|-------------|---------------|
| PanNuke   | Primary benchmarking | Multi-organ | Pickled bytes |
| PUMA      | TILs detection       | Melanoma    | TIFF, GeoJSON |
| panopTILs | TILs detection       | Breast      | PNG           |
| MoNuSAC   | TILs detection       | Multi-organ | Pickled bytes |

Table: Summary of datasets used in this study.

- H&E stained histopathology images
- $\sim 0.25$  MPP resolution
- Multiple tissue types for generalization

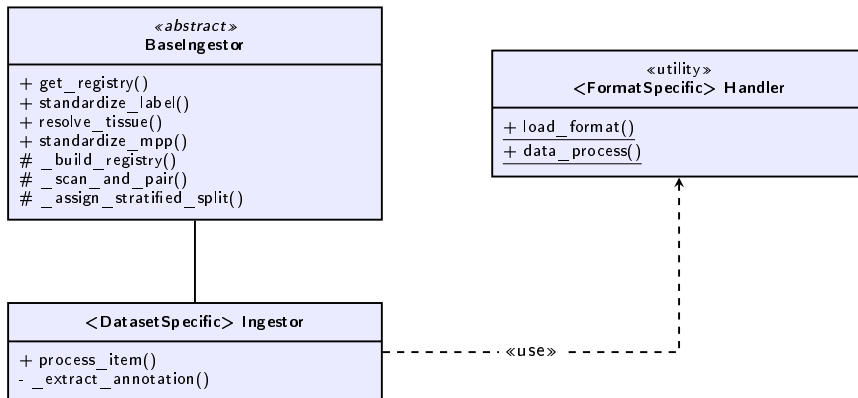


# Data Preprocessing Pipeline



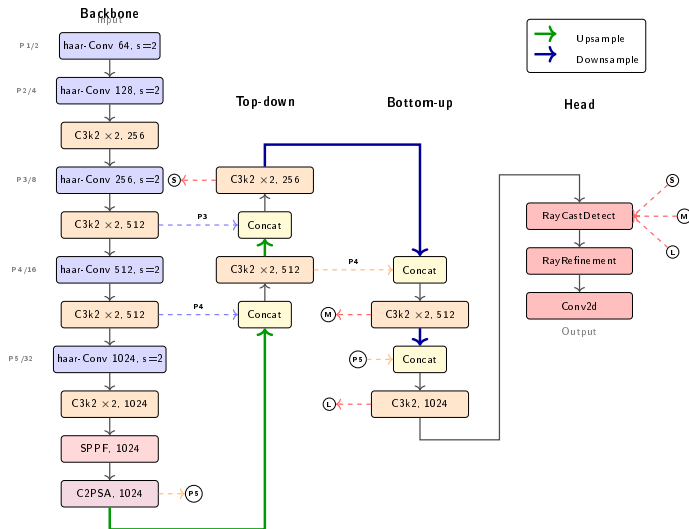
- Multi-format dataset handling (TIFF, PNG, pickled bytes)
- Patch extraction and augmentation
- Star-convex annotation generation

# Data Ingestion Design



- Multi-format dataset handling (TIFF, PNG, pickled bytes)
- Object-oriented design for modularity and extensibility

# RayCastED Architecture



# Key Innovations

## 1. RayCast Representation

- Centroid  $(x, y) + N$  radial distances
- Lightweight, boundary-aware representation

## 2. End-to-End Assignment

- Learned bipartite matching
- Eliminates NMS dependency

## 3. Lightweight FCN Backbone

- Efficient wavelet-enhanced feature extraction
- TensorRT/ONNX compatible

# Evaluation Procedure



# Evaluation Against LSP-DETR

Table: Evaluation Against LSP-DETR

| Metric         | LSP-DETR     | RayCastED    |
|----------------|--------------|--------------|
| F1             | <b>0.825</b> | 0.748        |
| AJI            | <b>0.677</b> | 0.546        |
| Precision      | <b>0.862</b> | 0.7618       |
| Recall         | <b>0.791</b> | 0.7348       |
| bPQ            | <b>0.657</b> | 0.5769       |
| mPQ            | <b>0.476</b> | 0.4381       |
| Params         | 45.0M        | <b>2.99M</b> |
| Inference Time | -            | 3.75ms       |